ORIENT: Technical Supplement Method Details, Protocols, and Additional Analyses

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This companion to *ORIENT: Online Reinforcement Learning for Data Selection in MLLM Instruction Tuning* provides further details on constrained subset inference, policy updates, data preparation, and experimental analyses. It provides complete signal-range scores, metric-omission results, and detailed grouping diagnostics. Equations, tables, and figures use independent numbering. Code is available at https://github.com/Meltcjm/ORIENT.

APPENDIX A
DATASETS, FEEDBACK SPLITS, AND CANDIDATE FEATURES

A. Training Collections and Feedback Holdout

The experiments use LLaVA-665K and the 186K training subset of Vision-Flan [1], with LLaVA-1.5-7B [2]. The larger-model experiment uses LLaVA-1.5-13B on LLaVA-665K. Dataset names identify the original collections. Candidate counts exclude the examples held out for feedback.

Before constructing the stage candidate pools, examples are randomly held out within each native task. The holdout rule allocates 23 examples to each of LLaVA-665K's 11 tasks, giving 253 feedback examples. The supplied Vision-Flan partition contains 191 native tasks; allocating two examples per task allocates 382 feedback examples under this rule (Table S1). The split manifest lists the individual examples assigned to each split in a run. The remaining examples form the candidate pool $\mathcal{D}$. Feedback examples are fixed throughout a run, excluded from candidate selection and supervised fine-tuning (SFT), and used only to compute rewards. Native task partitions also govern task-balanced feedback in the transfer experiment (Appendix E).

The sample-level split determines the denominator $N = |\mathcal{D}|$ used by the budget calculation. Online training also depends on the stage assigned to each selected example and the training order within that stage. Earlier selected subsets change the model used to score later candidates.

TABLE S1
FEEDBACK ALLOCATIONS SPECIFIED BY THE PER-TASK HOLDOUT RULE. TOTALS ARE DERIVED FROM THE LISTED TASK COUNTS.

Collection	Tasks	Per task	Total
LLaVA-665K	11	23	253
Vision-Flan (186K)	191	2	382

B. Stage Construction and Training Exposure

Let $N = |\mathcal{D}|$ after feedback exclusion, ρ be the nominal selection ratio, B the effective global batch size per SFT optimizer update, and n the number of updates in a full stage. The batch-aligned selection budgets are

$$K = B\left\lfloor \frac{N\rho}{B} \right\rfloor, \qquad K_t = \min(nB, K - tnB). \tag{S1}$$

For $K \geq B$, there are $H = \lceil K/(nB) \rceil$ stages. The candidate pool is shuffled once and partitioned into disjoint stage pools. Each candidate is considered only at its assigned stage. Unselected examples are not returned to later pools. Nonfinal stage t ends at shuffled position $\lceil (\sum_{s=0}^{t} K_s)/\rho \rceil$; the final stage receives the remaining candidates. Stage t performs K_t/B SFT updates, with model, optimizer, and scheduler states carried forward.

The main selection ratio is $\rho = 0.20$, the effective global batch size is $B = 64$, and the standard stage length is $n = 12$. Main runs use two GPUs, a per-device microbatch of 16, and two-step gradient accumulation. A full stage therefore selects 768 examples and performs 12 SFT updates. The total numbers of selected examples and updates depend on the candidate count after the feedback holdout. The one-GPU diagnostic in Appendix G uses its separately specified microbatch and accumulation settings.

C. Loss Channels and Five Signals

Only supervised response tokens contribute to an example's negative log-likelihood (NLL). In a multi-turn conversation, each response is conditioned on the preceding dialogue. Image-grounded samples are evaluated with the image retained (channel V) and removed (channel T), using the same supervised response tokens. The frozen reference ξ shares the target model's initialization checkpoint before multimodal instruction tuning; θ_t is the evolving model before stage t. The five raw signals are

$$\begin{aligned} d^c &= \ell^c(\theta_t) - \ell^c(\xi), \quad c \in \{T, V\}, \\ g(\theta) &= \ell^T(\theta) - \ell^V(\theta), \quad \theta \in \{\xi, \theta_t\}, \\ \delta &= d^V - d^T = g(\xi) - g(\theta_t). \end{aligned} \tag{S2}$$

Reference losses are cached once. The current model scores each stage's candidates before selection. The signals describe the candidates, while stage feedback provides the target for policy learning.

D. Percentiles, Missing Channels, and Metadata

For each applicable raw signal, we also compute a midrank percentile among candidates from the same task and stage. For a signal value v among m candidates of a task, the percentile is $(n_{<v} + n_{=v}/2)/m$. The two counts are the numbers of values below and equal to the given value, respectively. Tied values receive the same percentile, and a constant signal has percentile 0.5. Percentiles are recomputed as the candidate distribution changes, even though the raw signals in (S2) have algebraic dependencies.

Candidate features combine the five raw signals, their percentiles, and metadata for image availability, task identity, and dialogue length. Only d^T is measured for text-only examples. Their raw feature tuple, ordered as $(d^V, d^T, \delta, g(\xi), g(\theta_t))$, is encoded as $(0, d^T, -d^T, 0, 0)$; the four unavailable percentiles are set to 0.5. This convention preserves the Actor input dimension. The signal identity in (S2) applies to image-grounded examples.

At initialization, $\theta_0 = \xi$, so $d^T = d^V = \delta = 0$ and $g(\theta_0) = g(\xi)$ under identical scoring conditions. The constant relative-difficulty and gain-gap signals receive percentile 0.5 and are excluded from the low-range cost. Visual-contribution values may still vary across image-grounded candidates, and the Actor also receives metadata. Later SFT stages allow the current and reference losses to diverge.

E. Quotas and Content Clusters

At each stage, a task with m_τ candidates receives provisional allocation $q_\tau = K_t m_\tau / |\mathcal{P}_t|$. Set $k_\tau = \lfloor q_\tau \rfloor$ and assign the remaining $K_t - \sum_\tau k_\tau$ slots to tasks with the largest fractional remainders. This preserves $\sum_\tau k_\tau = K_t$ and each task's capacity. Within each task, let $\mathbf{h}_{i,t}$ be the normalized final-layer hidden state immediately before the first supervised answer token. It is obtained from the visual channel when an image is present and the text channel otherwise. Let p_i be the answer-token-length percentile within the task over the full candidate pool $\mathcal{D}$. Candidate distances are

$$\mathrm{dist}_t(i,j) = \|\mathbf{h}_{i,t} - \mathbf{h}_{j,t}\|_2^2 + (p_i - p_j)^2. \tag{S3}$$

Each candidate is linked to its nearest neighbor within the task; edges are treated as undirected. Connected components define the clusters, which are rebuilt at each stage without fixing the number of clusters in advance. A task with quota k and C clusters must cover exactly $\min(k, C)$ distinct clusters. The learned policy and the matched random control use the same set of feasible subsets, called the policy's support.

F. Image–Text Mismatch Construction

The mismatch experiment uses LLaVA-665K and LLaVA-1.5-7B with a 20% selection budget. After holding out feedback examples, we randomly select 10% or 20% of the image-grounded training examples, pair the selected examples, and swap images within each pair. Their instructions and responses remain unchanged. The feedback set is unaffected, and all methods receive the same corrupted data at each rate. Relative performance is normalized by the clean full-data reference.

G. Scope of Cached and Refreshed Quantities

Reference losses and full-pool answer-length percentiles remain fixed. Each new stage computes current-model losses and representations for its assigned candidates, then constructs stage-local signal percentiles, quotas, and clusters. During the subsequent proximal policy optimization (PPO) update, those features and constraints stay fixed. The Actor changes candidate scores and recomputes the exact distribution. The old and updated policies therefore use the same feasible subsets when computing the probability ratio and Kullback–Leibler (KL) divergence. In transfer, this refresh schedule continues with the target MLLM while the source policy parameters stay fixed.

APPENDIX B
EXACT CONSTRAINED SUBSET INFERENCE

This section derives the inference rules for both coverage cases, explains backward sampling, and derives the policy statistics used during learning. Fix a stage and task with m candidates, quota k, and C nonempty, disjoint clusters. Write $m_j = |\mathcal{C}_j|$, $w_i = \exp(z_i) > 0$, and

$$\Omega = \{S : |S| = k,\ |\{j : S \cap \mathcal{C}_j \neq \varnothing\}| = \min(k, C)\}. \tag{S4}$$

For $S \in \Omega$, its probability is $\pi(S) = \prod_{i \in S} w_i / Z$, where Z sums these weights over Ω.

A. Coverage Cases and Feasibility

For $0 \le k \le m$, maximum coverage is achievable. If $k \le C$, choose k clusters and one candidate from each. If $k > C$, first choose one candidate from every cluster and distribute the remaining $k - C$ selections among their remaining capacities. The clusters have $m - C \ge k - C$ candidates left, so they can accommodate all remaining selections.

Thus, with $a_j = |S \cap \mathcal{C}_j|$, the feasible count sets are

$$\mathcal{A}_j = \begin{cases} \{0, 1\}, & k \le C, \\ \{1, \ldots, \min(m_j, k - C + 1)\}, & k > C, \end{cases} \tag{S5}$$

together with $\sum_j a_j = k$. At $k = C$, every cluster contributes exactly one candidate. A zero quota gives the unique empty subset, $Z = 1$, and log-probability zero. At $k = m$, the whole task pool is the unique action. In these deterministic cases, entropy is zero and inclusion probabilities do not change with the scores. A task with one candidate forms a cluster containing only that candidate.

B. Polynomial Form and Dynamic Programming (DP)

Let $W_{j,q}(a)$ sum the weights of size-a subsets of the first q candidates in cluster j. Partitioning subsets by whether they contain candidate q gives

$$W_{j,q}(a) = W_{j,q-1}(a) + w_{j,q} W_{j,q-1}(a-1). \tag{S6}$$

The boundary values are $W_{j,0}(0) = 1$ and zero for invalid counts. Set $W_j(a) = W_{j,m_j}(a)$ and define

$$P_j(x) = \sum_{a \in \mathcal{A}_j} W_j(a) x^a. \tag{S7}$$

Then $Z = [x^k] \prod_{j=1}^{C} P_j(x)$, where $[x^k]$ extracts the coefficient of x^k. Each product term picks one allowed subset from each disjoint cluster. Its degree is the total selected count, and its coefficient is the product of candidate weights. Consequently, the degree-k coefficient contains every feasible subset exactly once.

Writing $Z_j(s) = [x^s] \prod_{u=1}^{j} P_u(x)$ yields

$$Z_j(s) = \sum_{a \in \mathcal{A}_j,\ a \le s} W_j(a) Z_{j-1}(s-a), \tag{S8}$$

with $Z_0(0) = 1$ and zero elsewhere. The desired normalizer is $Z_C(k)$.

For $k \le C$, define $M_j = \sum_{i \in \mathcal{C}_j} w_i$. Each cluster polynomial reduces to $1 + M_j x$, giving

$$Z_j(s) = Z_{j-1}(s) + M_j Z_{j-1}(s-1). \tag{S9}$$

For $k > C$, let $r = k - C$ count selections beyond the mandatory one per cluster. Factor $P_j(x) = xQ_j(x)$, where

$$Q_j(x) = \sum_{u=0}^{\min(m_j-1, r)} W_j(u+1) x^u. \tag{S10}$$

The normalizer is the degree-r coefficient of $\prod_j Q_j(x)$. Factoring out the mandatory selections lets the DP track only the extra selections.

C. Backward Sampling and Its Distribution

At state (j, s) of the DP over clusters, sample the count for cluster j with probability

$$\Pr(a \mid j, s) = \frac{W_j(a) Z_{j-1}(s-a)}{Z_j(s)}, \tag{S11}$$

then continue from $(j-1, s-a)$. Equation (S8) makes these probabilities sum to one. Within cluster j, given a remaining selections among the first q candidates, include candidate q with probability

$$\frac{w_{j,q} W_{j,q-1}(a-1)}{W_{j,q}(a)}. \tag{S12}$$

Decrease a after inclusion and decrease q at every step. At $a = 0$, all remaining candidates are excluded; at $a = q$, all are included.

Multiplying the probabilities of the cluster-count decisions cancels the intermediate DP terms, leaving $\prod_j W_j(a_j)/Z$. The same cancellation within each cluster gives $\prod_{i \in S \cap \mathcal{C}_j} w_i / W_j(a_j)$. Multiplying these expressions cancels every $W_j(a_j)$ and gives

$$\Pr(\text{sampler returns } S) = \frac{\prod_{i \in S} w_i}{Z}. \tag{S13}$$

Every sampled action therefore follows the specified subset distribution. Processing candidates in a fixed reverse order gives each subset a unique decision path.

D. Inclusion Probabilities and Policy Gradients

Let $I_i(S) = \mathbf{1}[i \in S]$. Differentiating the finite sum defining Z gives

$$\mu_i = \frac{\partial \log Z}{\partial z_i} = \sum_{S \in \Omega} \pi(S) I_i(S). \tag{S14}$$

It follows that $0 \le \mu_i \le 1$, $\sum_i \mu_i = k$, and

$$\frac{\partial \log \pi(S)}{\partial z_i} = I_i(S) - \mu_i. \tag{S15}$$

A second derivative gives the covariance matrix of selection indicators:

$$\frac{\partial \mu_i}{\partial z_l} = \mathbb{E}[I_i I_l] - \mu_i \mu_l. \tag{S16}$$

This identity shows how changing one candidate's score affects inclusion probabilities under the quota and coverage constraints.

Let $\mathcal{V}_i$ index candidate i's applicable, nonconstant signals, and $u_{i,v}$ their task-relative percentiles. The fixed candidate cost is

$$a_i = \begin{cases} |\mathcal{V}_i|^{-1} \sum_{v \in \mathcal{V}_i} \mathbf{1}[u_{i,v} < 0.1], & |\mathcal{V}_i| > 0, \\ 0, & |\mathcal{V}_i| = 0. \end{cases} \tag{S17}$$

The stage penalty satisfies

$$\mathcal{L}_{\text{low},t} = \mathbb{E}\left[\frac{1}{K_t} \sum_{i \in S_t} a_i\right] = \frac{1}{K_t} \sum_i a_i \mu_i. \tag{S18}$$

For a candidate l in a task, its score derivative is

$$\frac{\partial \mathcal{L}_{\text{low},t}}{\partial z_l} = \frac{1}{K_t} \sum_{i \text{ in the same task}} a_i \operatorname{Cov}(I_i, I_l). \tag{S19}$$

The threshold acts on task-relative percentiles. A percentile equal to 0.1 incurs no penalty. Constant signals and unavailable visual signals are excluded from both the count and the average. The cutoff is separate from the PPO clipping width $\epsilon = 0.2$.

E. Entropy, Exact KL, and Uniform Feasible Selection

Taking expectations of $-\log \pi(S)$ yields

$$\mathcal{H}(\pi) = \log Z - \boldsymbol{\mu}^\top \mathbf{z}. \tag{S20}$$

For old scores $\mathbf{z}^-$ and updated scores $\mathbf{z}$ on the same feasible support,

$$\begin{aligned} D_{\mathrm{KL}}(\pi^- \| \pi) = {} & \log Z(\mathbf{z}) - \log Z(\mathbf{z}^-) \\ & - \boldsymbol{\mu}(\mathbf{z}^-)^\top (\mathbf{z} - \mathbf{z}^-). \end{aligned} \tag{S21}$$

The stage distribution is the product of the task distributions. Its log-probability, entropy, and KL divergence are therefore the sums of the corresponding task quantities. The Actor entropy term divides the summed entropy by K_t; the KL limit applies to the summed full-stage divergence.

Setting every score to zero gives $Z = |\Omega|$ and $\pi(S) = 1/|\Omega|$. Candidate inclusion probabilities, or marginals, still depend on the constraints. For example, consider clusters $\{x_1, x_2\}$ and $\{x_3\}$ with quota two. The two feasible actions are $\{x_1, x_3\}$ and $\{x_2, x_3\}$, each with probability $1/2$. The marginals are $(1/2, 1/2, 1)$. This is the baseline used in the enrichment denominator in Appendix F.

F. Numerical Evaluation and Complexity

The polynomial recurrences can be evaluated in log space. Products become sums, and log-sum-exp combines alternative branches. Unreachable states have log-weight $-\infty$. For any common task-level shift c, replacing z_i by $z_i - c$ changes $\log Z$ by $-kc$ and preserves probabilities, marginals, entropy, and KL. These identities permit score centering before inference. The inspected reference implementation uses double-precision scores, task-wise mean centering, and log-domain coefficient updates.

For $k \le C$, computing the cluster weight sums costs $O(m)$ and the cluster DP costs $O(Ck)$. For $k > C$, the local coefficients up to degree $r+1$ cost $O(m(r+1))$. A cluster contributes at most $\min(m_j, r+1)$ shifted coefficients; combining it with at most $r+1$ prefix states costs $O((r+1)\min(m_j, r+1))$. Summing over clusters gives $O(m(r+1))$. Thus

$$D = \begin{cases} O(m + Ck), & k \le C, \\ O(m(k - C + 1)), & k > C. \end{cases} \tag{S22}$$

Stored intermediate DP values support backward sampling and differentiation within the same asymptotic time bound. Computing the low-range gradient differentiates through the marginals in (S14). With a bounded number of Actor steps per stage, these operations add constant factors to D.

Clustering has a separate cost for each task of $O(d_h m^2)$ for d_h-dimensional representations. If stage sizes remain bounded as the dataset grows, summing the stage costs is linear in candidate count. Computing full-pool answer-length percentiles once contributes $O(N \log N)$ preprocessing. The total bound therefore retains this preprocessing term and the stage-local quadratic clustering term.

APPENDIX C
TRAINING CONFIGURATION AND POLICY UPDATES

A. SFT and Policy Networks

SFT uses the LLaVA-1.5 low-rank adaptation (LoRA) hyperparameters [2] listed in Table S2, with an effective global batch size of 64. For each model size, the target model θ_0 and reference ξ start from the same initialization checkpoint before multimodal instruction tuning; ξ remains frozen. Policy-network settings appear in Table S3.

For W devices, per-device microbatch b, and accumulation factor a, $B = baW$. Main experiments use $W = 2$, $b = 16$, and $a = 2$, giving $B = 64$. The one-epoch setting applies to the selected examples. Appendix A specifies how many examples and updates online training uses. Stage-wise reward evaluation is performed by the selection loop.

The Critic receives training progress t/H, the preceding reward, the mean and population standard deviation of the most recent eight completed rewards, and $\log(1+t)$. The Critic uses training history as input, while the Actor receives candidate features. The two networks have separate trainable parameters.

TABLE S2
SFT SETTINGS CONFIRMED FROM THE SUPPLIED LLAVA-1.5 LORA SCRIPT, WITH THE GLOBAL BATCH SIZE ADJUSTED FOR OUR EXPERIMENTS.

Setting	Value
LoRA rank, alpha	128, 256
SFT learning rate	2×10^{-4}
Projector learning rate	2×10^{-5}
Weight decay	0
Scheduler, warmup ratio	Cosine, 0.03
Effective global training batch	64
Per-device batch, accumulation	16, 2 (two GPUs)
Epoch setting	1
Maximum sequence length	2,048
BF16, TF32	Enabled, enabled
Gradient checkpointing	Enabled
Distributed configuration	DeepSpeed; `zero3.json`
Group by modality length	Enabled
Vision tower	CLIP ViT-L/14, 336-pixel input
Vision feature layer	-2
Projector type	`mlp2x_gelu`
Image aspect handling	Padding
Image start, end, patch tokens	All disabled
Conversation version	`v1`
Data-loader workers, lazy preprocessing	4, enabled

TABLE S3
SPECIFIED SETTINGS FOR THE STANDARD ONLINE POLICY.

Setting	Value
Actor hidden layers	256, 128, 64; GELU
Critic hidden layers	32, 32; Tanh
Actor, Critic optimizers	AdamW, AdamW
Actor, Critic learning rates	3×10^{-4}, 3×10^{-4}
PPO calls per completed transition	1
Actor, Critic steps per call	At most 2, 1
Discount γ	0.95
PPO clip width ϵ	0.2
Critic loss weight	0.5
Entropy weight	0.01
Low-range penalty weight	1
Low-range percentile cutoff	0.1
Exact full-set KL limit	0.03
Gradient norm clipping	0.5 for each network
Standard selection ratio, stage length	20%, 12 SFT updates
Main hardware	Two A100-SXM4-80GB GPUs

B. Feedback, Transition Timing, and Bootstrapping

We first average the loss over the fixed feedback examples in each task, then average the task losses with equal weight:

$$L_F(\theta) = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \frac{1}{|F_\tau|} \sum_{x \in F_\tau} \ell(\theta; x), \qquad r_t = L_F(\theta_t) - L_F(\theta_{t+1}). \tag{S23}$$

Feedback examples retain their original inputs, including images when present. The post-SFT loss of one stage serves as the pre-SFT loss of the next stage.

TABLE S4
ORDERING AROUND A NONTERMINAL STAGE TRANSITION.

Order	Operation
1	Sample S_t; cache sampling scores, log-probability, and Critic value.
2	Train on S_t for K_t/B updates; measure r_t.
3	Prepare the next stage's features, constraints, and history.
4	Form $\widehat{R}_t$ and fixed advantage using the pre-update Critic.
5	Optimize the Actor at most twice, checking each step; update the Critic once.
6	Select S_{t+1} using the updated Actor and prepared next-stage inputs.

At selection time, the policy inputs, sampled subset log-probability, and Critic value are cached. Each stage collects feedback once after its SFT updates and produces one transition for one PPO call. For a transition before the final stage, we prepare the next observation, complete the PPO call, and then select the next subset. The return target uses the Critic before that update:

$$\widehat{R}_t = r_t + \gamma V_{\psi^-}(\mathbf{b}_{t+1}), \qquad t < H-1. \tag{S24}$$

The final target is $\widehat{R}_{H-1} = r_{H-1}$. The advantage is the return target minus the value cached when the action was sampled, with no additional scaling. Each PPO call performs at most two Actor optimizer steps and one Critic optimizer step on the same transition. At $n = 12$, each full stage therefore contributes 12 SFT updates followed by feedback and a PPO call. The final, possibly shorter stage follows the same feedback rule.

Both Actor steps use the same cached sampling probability in the ratio denominator. In log form,

$$\omega_t(\phi) = \exp\big(\log \pi_\phi(S_t) - \log \pi_{\phi^-}(S_t)\big). \tag{S25}$$

The ratio starts at one. If the first step is accepted, the second step uses the updated numerator and keeps the denominator and advantage fixed. Clipping then applies at $1 \pm \epsilon$. Both steps hold candidate features, quotas, cluster assignments, and per-candidate low-range costs fixed.

Writing $L_t = L_F(\theta_t)$, the discounted objective can also be expanded as

$$\sum_{t=0}^{H-1} \gamma^t r_t = L_0 - (1-\gamma) \sum_{t=1}^{H-1} \gamma^{t-1} L_t - \gamma^{H-1} L_H. \tag{S26}$$

At the reported $\gamma = 0.95$, it rewards lower intermediate and final feedback losses. At $\gamma = 1$, it reduces to $L_0 - L_H$.

C. Entropy, the Low-Range Penalty, and KL Control

For each candidate, compute the fraction a_i of its applicable, nonconstant signal percentiles strictly below 0.1, or zero if none are applicable (Eq. (S17)). Image-grounded candidates may contribute all five signals; text-only candidates contribute d^T. The stage penalty is the exact expectation of the mean cost over selected examples,

$$\mathcal{L}_{\text{low},t} = \frac{1}{K_t} \sum_{i \in \mathcal{P}_t} a_i \mu_{i,t}, \tag{S27}$$

where $\mu_{i,t}$ is the inclusion probability under the constrained policy. During an update, a_i, the features, task quotas, and clusters are fixed. Gradients flow through $\mu_{i,t}$. The penalty includes applicable signals even when they are correlated.

For a task policy with candidate scores $\mathbf{z}$, inclusion probabilities $\boldsymbol{\mu}$, and normalizer Z, its entropy is $\log Z - \boldsymbol{\mu}^\top \mathbf{z}$. The sum across tasks is divided by K_t to obtain the entropy term. The exact KL divergence for the full selection policy is the sum of the divergences for individual tasks. Candidate features and feasible sets stay fixed when comparing old and updated policies. After each Actor step, the cumulative KL is checked against the original sampling policy, with limit 0.03. If a step exceeds the limit, the Actor parameters are restored to their values before that step. If the second step is rejected, the accepted first step is retained. The original sampling policy remains the reference for both checks.

APPENDIX D
ABLATION AND RANDOM-CONTROL DEFINITIONS

At matched selection budgets, the compared methods train on the same proportion of the candidate pool, use the same SFT hyperparameters, and are evaluated on the same benchmark suite. The full-data model provides a separate normalization reference. All component comparisons use the LLaVA-665K pool, 20% budget, task quotas, stage length $n = 12$, initialization, SFT updates, and evaluation protocol specified in the manuscript. Table S5 distinguishes the changed mechanisms.

A. Removing Reward-Driven Actor Updates

The w/o R Actor objective is

$$\mathcal{L}^{\text{w/o R}}_{\text{Actor},t} = -\beta_{\text{ent}} \text{Ent}_t + \mathcal{L}_{\text{low},t}. \tag{S28}$$

The Critic continues to update against its return targets. With separate network parameters, Actor gradients come entirely from entropy and the low-range penalty. Relative performance falls from 98.70% to 96.29% in this comparison. The 2.41-percentage-point difference supports the benefit of reward-driven policy updates when the regularizers are retained.

B. Uniform Sampling from the Same Feasible Sets

Let $\Omega_{t,\tau}$ contain the feasible subsets for task τ at stage t. Its policy assigns a feasible subset S weight $\exp\big(\sum_{i \in S} z_{i,t}\big)$. In the matched random control, all $z_{i,t}$ are zero. Every feasible subset then has weight one, so

$$\begin{aligned} Z_{t,\tau} &= |\Omega_{t,\tau}|, \\ \pi_{\text{rand},\tau}(S) &= \frac{1}{|\Omega_{t,\tau}|}, \quad S \in \Omega_{t,\tau}. \end{aligned} \tag{S29}$$

Tasks have fixed quotas, and each feasible stage subset combines one feasible subset from each task. Sampling independently and uniformly within each task therefore gives a uniform distribution over all feasible stage subsets. Zero quotas produce the unique empty subset for the affected task.

This control uses the DP recurrences and backward sampler in Appendix B. It reaches 94.94% Rel., compared with ORIENT's 98.70%. The 3.76-point difference measures the benefit

TABLE S5
MECHANISMS CHANGED IN THE REPORTED CONTROLS. THE MATCHED RANDOM CONTROL DIFFERS FROM THE ORDINARY RANDOM BASELINE IN THE MAIN COMPARISON TABLES.

Variant	Changed component	Retained mechanisms, interpretation
w/o R	Remove the Actor's reward-driven PPO term.	Retain entropy, the low-range penalty, refreshed features, and KL control. Critic updates continue but cannot affect Actor optimization.
w/o C	Remove within-task cluster coverage.	Retain task quotas, reward learning, and the low-range penalty.
w/o T	Remove the low-range penalty.	Retain reward learning, entropy, task quotas, cluster coverage, and KL control.
Sequential policy (Seq.)	Sample without replacement using softmax probabilities from Actor scores, excluding choices that make the constraints impossible to satisfy.	Retain task quotas, cluster coverage, stage rewards, and Critic-based advantage estimation. PPO uses the sampled sequence probability.
Matched Random	Set every candidate score to zero in the same exact DP sampler; perform no reward learning or low-range penalty optimization.	Retain the same candidates, task quotas, and cluster-coverage constraints; every feasible subset has equal probability.

of the full policy and its training procedure over uniform feasible selection; w/o R separately tests reward-driven Actor updates.

C. Sequential Policy Comparison

Given the candidates already selected, $s_{<l}$, let $\mathcal{E}_l$ contain the candidates that can still be added while allowing the final subset to satisfy the task quotas and maximum coverage. The conditional rule and sampled-sequence log-probability are

$$\begin{aligned} p(i_l \mid s_{<l}) &= \frac{\exp(z_{i_l})}{\sum_{u\in\mathcal{E}_l}\exp(z_u)}, \quad i_l \in \mathcal{E}_l, \\ \log p(i_1,\ldots,i_{K_t}) &= \sum_{l=1}^{K_t} \log p(i_l \mid s_{<l}). \end{aligned} \tag{S30}$$

PPO uses this sequence probability with the same stage reward and Critic-based advantage. Seq. reaches 97.74% Rel., with SQA-I and GQA each 0.10 above the full method and ten metrics below it. The comparison evaluates the resulting constrained selection policies.

APPENDIX E
FROZEN SOURCE-TO-TARGET POLICY TRANSFER

A. Transfer Interface

The reported transfer setting loads the Actor and Critic trained on LLaVA-665K, freezes their parameters, and uses the Actor to select 20% of Vision-Flan. The target model continues SFT on the examples selected at each target stage. The source Actor's task input has 11 coordinates, corresponding to ten image-grounded groups and one text-only group.

Let $\mathcal{T}_{\mathrm{VF}}$ denote the native target tasks and $\mathcal{T}_{\mathrm{src}}$ the source task identities. A fixed map $h : \mathcal{T}_{\mathrm{VF}} \to \mathcal{T}_{\mathrm{src}}$ supplies the task identity used by the Actor. Let $\iota(s)$ be the coordinate assigned to source task s in the saved policy checkpoint. A candidate from native task τ receives

$$\mathbf{u}(\tau) = \mathbf{e}_{\iota(h(\tau))} \in \{0,1\}^{11}. \tag{S31}$$

Native target labels determine quotas, signal percentiles, content clusters, and task-balanced feedback. Several native tasks can share an Actor identity while retaining separate selection constraints.

TABLE S6
TASK IDENTITY AND MODEL-STATE HANDLING DURING TRANSFER.

Quantity	Treatment on Vision-Flan
Actor task input	11-dimensional source identity through h
Task quotas	Native target tasks
Signal percentiles	Native task and current stage
Content clusters	Native task and current target-model representation
Feedback averaging	Native target tasks
Source Actor, Critic	Parameters frozen
Target MLLM	Updated by SFT
Reference model and losses	Frozen initialization; target reference losses cached
Current-model signals	Recomputed before target-stage selection

As target SFT progresses, both the numerical signals and their task-relative percentiles can change. The frozen Actor applies its learned scoring function to these refreshed inputs, and the exact sampler enforces target-stage constraints. Thus, the selection distribution changes as its inputs and feasible subsets change, even though the policy parameters remain fixed.

The main-paper comparison reports 102.9% Rel. and 96.73% minimum full-data retention for the transfer setting; the target-trained policy reaches 103.0% Rel. The following subsections document the task mapping used in this transfer experiment.

B. Semantic Task Mapping

The mapping uses source and target instructions and response formats. The supplied Vision-Flan partition contains 191 task files and 186,103 image-grounded records. Its task-name list and aggregate dictionary match these files. The ten supplied LLaVA image-grounded groups contain 624,610 records. The source identities also include a text-only group. Table S7 lists the source groups and mapped target counts.

For each native task, we assign one source identity based on its requested operation, answer format, and required visual

evidence. All its examples share this identity. We apply these rules in order.

1) **Modality and region coordinates.** Text-only instructions use the text-only identity. Tasks that convert region coordinates to descriptions, or descriptions to coordinates, use a corresponding grounding identity. Object names, category lists, and counts follow the recognition rules below.
2) **Indexed answer selection.** Tasks that return a letter, a numbered candidate, or a question–answer index pair use A-OKVQA. Direct category labels follow the recognition rule; direct spatial or temporal relation answers follow the relational rule.
3) **Dialogue and generation.** Continuing an explicit dialogue uses the conversation identity. Explanations, rationales, background knowledge, and open-ended reasoning about procedures use reasoning. Captions, object descriptions, and descriptions of visible changes use description.
4) **Text-dependent evidence.** Transcription and short answers grounded in image text, documents, or chart values use OCR-VQA. Descriptions that integrate image text or chart values use TextCaps. This rule also applies to captions that depend on image text.
5) **Direct and relational answers.** Direct recognition, attributes, category labels, and simple counts use VQAv2. Spatial relations, ordering, comparisons, compositional conditions, and structured numerical reasoning about visual content use GQA.

Source examples help distinguish overlapping functions. VQAv2 includes website-identification questions, supporting its use for CoVA webpage recognition. TextCaps includes chart and diagram descriptions, supporting its use for Chart2Text summaries. Source VQAv2 and GQA records both contain multiple short question–answer exchanges about an image; the requested operation and answer content determine their assignments.

C. Task Names and Boundary Cases

Task names may not reflect the requested operation. *MEMOTION+sentiment_detection* asks for image-text transcription and maps to OCR-VQA. *A-OKVQA+answer_rationales_matching* requests an answer with an explanation and maps to reasoning. *Winoground+image_captioning* selects between two indexed captions and maps to A-OKVQA. *FFHQ-Text+text-to-face_generation* takes a face image and produces a description.

The CLEVR context variants provide preceding question–answer facts and end with a compositional visual question, so they map to GQA. The two WikiHow step-order tasks return *next* or *previous*; they use GQA's relational identity. Spot-the-Diff produces short descriptions of changes between two panels and maps to description. The target *visualgenome_vqa* task returns short visual answers and maps to VQAv2; source group L07 uses region-coordinate interfaces.

The records identify 152 tasks with direct functional matches, 30 with overlapping source functions, and nine with approximate matches. The nine approximate cases are Chart2Text, VQG question generation, Geometry3K, RAVEN, Recipe-QA visual coherence, VQA-RAD, WikiHow next-step generation, and the two WikiHow step-order tasks. These tasks involve chart summaries, question generation, specialist reasoning, medical images, or procedural relations beyond the closest source task's usual requirements. Each has a fixed assignment and explicit alternatives in the accompanying manifest. For example, VQG question generation uses reasoning's open-ended generation identity, with conversation and description recorded as alternatives.

Eight source identities receive target tasks. The two region-coordinate identities and the text-only identity retain their source coordinates, but their entries are zero for every target example. Task counts in Table S7 follow the semantic assignments, while target quotas continue to use the 191 native partitions.

D. Feature Construction and Checkpoint Binding

The adapter first constructs candidate features using native target labels, including the task-local signal percentiles and the Low-Range Penalty with threshold 0.1. It then replaces the task one-hot block using (S31). In the inspected specification, the Actor input has 24 entries: five raw signals, an image-availability flag, two features for prompt length and turn count, 11 source-task coordinates, and five native-task signal percentiles.

To match the inputs to the checkpoint, the adapter reads the saved source task names and feature order. It preserves the source weights and normalization buffers, freezes the Actor and Critic, and uses the native target labels for all constraint and feedback calculations. During frozen-policy transfer, each stage refreshes target-model features and continues SFT without updating the policy.

The companion adapter checks task-name coverage, source identities, feature order, and file hashes against the specification. Its checks cover all 191 assignments, different source-coordinate orders, unknown tasks, duplicate labels, and preservation of the raw-feature and native-percentile blocks. They also verify that replacing task identities leaves the Low-Range Penalty and native-task inputs unchanged.

E. Complete Assignment and Mapping Records

Tables S8–S10 list all 191 native tasks and their source-group codes. The companion JSON and CSV record exact task names, record counts, instruction and response excerpts, sample IDs, source examples, mapping reasons, alternatives, and target-file SHA-256 hashes. Mapping decisions use task data and source examples; feedback and benchmark scores are excluded. Each run records the mapping-file hash, source checkpoint, and native split manifest alongside its selections.

TABLE S7

SOURCE IDENTITIES AND TARGET COVERAGE OF THE MAPPING USED FOR POLICY TRANSFER. COUNTS REFER TO THE SUPPLIED TARGET FILES BEFORE FEEDBACK EXCLUSION. GROUP CODES IDENTIFY SEMANTIC GROUPS; CHECKPOINT METADATA DETERMINES THEIR ONE-HOT COORDINATES.

Code	Source identity	Function	Target tasks	Target records
L01	conversation	Visual dialogue	4	4,000
L02	description	Image and visible-change description	22	21,753
L03	reasoning	Explanations, background knowledge, and open generation	8	7,998
L04	vqav2	Short visual answers and direct recognition	89	85,910
L05	aokvqa	Indexed answer selection and matching	41	40,743
L06	refcoco	COCO region-coordinate grounding	0	0
L07	visual_genome	VG region-coordinate grounding	0	0
L08	gqa	Relational, compositional, and structured visual QA	13	13,000
L09	ocr_vqa	Text reading and document QA	12	10,699
L10	textcaps	Descriptions integrating image text or chart values	2	2,000
L11	text_only	Text-only instruction following	0	0
Total			191	186,103

TABLE S8

COMPLETE VISION-FLAN TO LLAVA TASK ASSIGNMENT (1/3). SOURCE CODES ARE DEFINED IN TABLE S7.

Native Vision-Flan task	Source	Native Vision-Flan task	Source
300w+human_portrait_classification	L04	CONCADIA+image_description	L02
A-OKVQA+answer_rationales_matching	L03	ConceptualCaptions+image_captioning	L02
A-OKVQA+rationales_generation	L03	Core50+Object_detection	L04
A-OKVQA+visual_question_answering	L04	CoVA+webpage_recognition	L04
AI2D+visual_question_answering	L05	crowdhuman+count_numbers	L04
AID+aerial_scene_classification	L04	CUB-200-2011+Bird_Classification	L04
ayahoo_test_images+animal_object_vehicle_image_classification	L04	DAQUAR+object_question_answer	L04
Caltech-256+image_classification	L04	Dark-Zurich+time_classification	L04
Caltech101+Image_classification	L04	DeepFashion_highres_Attribute_and_Category+Cloth_Classification	L04
Caltech101+Living_Thing_classification	L05	DeepWeeds+weed_species_recognition	L04
Cars+car_brand_classification	L04	DOCVQA+question_answer	L09
Cars+car_classification	L04	DOMAIN_NET+clipart_image_classification	L04
CHART2TEXT+chart_caption	L10	DOMAIN_NET+image_category_classification	L04
cinic-10+image_classification_animal	L04	DOMAIN_NET+infograph_image_classification	L04
cinic-10+image_classification_shipping	L04	DOMAIN_NET+painting_image_classification	L04
cinic-10+image_classification_transport	L04	DOMAIN_NET+quickdraw_image_classification	L04
cinic-10+object_presence_animal	L04	DOMAIN_NET+real_image_classification	L04
cinic-10+object_presence_shipping	L04	DTD+all_texture_detection	L04
Clevr+Multiple_Question_Answering	L08	DTD+coarse_grained_texture_classification	L04
CLEVR+question_answer	L08	DVQA+charts_question_answer_1	L09
Clevr+Question_Answer_Matching	L05	DVQA+charts_question_answer_2	L09
Clevr+Question_Answering	L08	ExDark+object_recognition	L04
Clevr+VQA_context	L08	expw+expression_detection	L04
CLEVR_CoGenT+Multiple_Question_Answering	L08	fairface+image_classification_age	L04
CLEVR_CoGenT+Question_Answer_Matching	L05	fairface+image_classification_gender	L04
CLEVR_CoGenT+Question_Answering	L08	fairface+image_classification_race	L04
CLEVR_CoGenT+VQA_context	L08	FFHQ-Text+text-to-face_generation	L02
coco+image_classification_animal	L05	FGVC_Aircraft+Aircraft_Classification	L04
coco+image_classification_appliance	L05	FGVC_Aircraft+Aircraft_Classification_Family	L04
coco+image_classification_furniture	L05	FGVC_Aircraft+Aircraft_Classification_Manufacturer	L04
coco+image_classification_kitchen	L05	FGVC_Aircraft+Aircraft_Classification_Variant	L04
coco+image_classification_sports	L05	FLICKR30K+caption_image	L02
coco+image_classification_vehicle	L05	FlickrLogos-27+logo_detection	L04
CONCADIA+image_caption_context_1	L03	FoodLogoDet-1500+food_logo_recognition	L04
CONCADIA+image_caption_context_2	L03	FUNSD+text_detection	L09

TABLE S9
COMPLETE VISION-FLAN TO LLAVA TASK ASSIGNMENT (2/3). SOURCE CODES ARE DEFINED IN TABLE S7.

Native Vision-Flan task	Source	Native Vision-Flan task	Source
GEOMETRY3K+geometry_question_answer	L05	MemeCap+meme_captioning	L03
GQA	L08	MEMOTION+sentiment_detection	L09
GTSRB+image_classification	L05	MNIST-M+number_recognition	L09
HICO+human_activity_detection	L04	model-vs-human+image_style_classification	L04
HICO+object_classification	L04	multimodal_factual_checking	L05
iconqa+choose_txt	L08	MVTecAD+anomaly_detection	L04
iconqa+fill_in_blank	L08	MVTecAD+image_classification	L04
image_caption	L02	NOCAPS+image_caption	L02
image_quality	L05	NUS-WIDE+Animal_classification	L05
image_text_selection	L05	ObjectNet+Object_classfication	L04
ImageNet-A+image_classification	L05	Office-Home+Image_classification	L04
ImageNet-C+image_classification_blur	L05	Office_31+Image_Classification_Category	L05
ImageNet-C+image_classification_general	L05	Office_31+Image_Classification_Object	L04
ImageNet-C+image_classification_noise	L05	Office_31+Image_Classification_ObjectAndCategory	L04
ImageNet-C+image_classification_weather	L05	ok_vqa	L04
ImageNet-R+image_classification	L04	PACS+art_painting_object_classification	L04
ImageNet-R+image_domain_classification	L04	PACS+cartoon_object_classification	L04
ImageNet-Sketch+image_classification	L04	PACS+dog_image_category_classification	L05
infographicvqa+question_answer	L09	PACS+elephant_image_category_classification	L05
infographicvqa+single_document_question	L09	PACS+giraffe_image_category_classification	L05
ITM	L05	PACS+guitar_image_category_classification	L05
KVQA+image_captioning	L02	PACS+horse_image_category_classification	L05
KVQA+image_question_answer	L04	PACS+house_image_category_classification	L05
LAD+object_detection_details	L02	PACS+person_image_category_classification	L04
LFW+face_recognition	L04	PACS+photo_object_classification	L04
LOC_NARRATIVES+ade20k_images_caption	L02	PICKAPIC+image_short_description	L02
LOC_NARRATIVES+coco_images_caption	L02	Places205+Image_env_classification	L05
LOC_NARRATIVES+flickr30k_images_caption	L02	places365+Image_Classification	L04
LOC_NARRATIVES+open_images_caption	L02	PlotQA+visual_question_answering	L09
LSUN+Image_Classification	L05	question_image_match	L05
MemeCap+image_captioning	L02		

TABLE S10
COMPLETE VISION-FLAN TO LLAVA TASK ASSIGNMENT (3/3). SOURCE CODES ARE DEFINED IN TABLE S7.

Native Vision-Flan task	Source	Native Vision-Flan task	Source
RAVEN+next_pattern	L05	visualgenome_vqa	L04
recipe-qa+visual_coherence	L05	VIZWIZ+image_captioning	L02
REDCAPS+reddit_caption_1	L02	vizwiz+question_answer	L05
REDCAPS+reddit_caption_2	L02	VOC2007+object_detection	L04
Road-Anomaly+anomaly_detection	L04	VQA	L05
SCUT-CTW1500+text_detection	L09	VQA-E+image_captioning	L02
semart+image_description	L02	VQA-E+visual_question_answering	L04
semart+image_school	L04	VQA_activity_recognition	L04
semart+image_technique	L04	VQA_attribute	L04
semart+image_timeframe	L04	VQA_color	L04
semart+image_type	L04	VQA_counting	L04
SentiCap+image_sentiment_captioning	L02	VQA_object_presence	L04
Set5+image_recognition	L04	VQA_object_recognition	L04
SKETCH+living_organism_detection	L04	VQA_positional_reasoning	L08
SKETCH+object_detection	L04	VQA_scene_recognition	L04
spot-the-diff+image_diff_identification	L02	VQA_sentiment_understanding	L04
STANFORD_DOGS+dog_classification	L04	VQA_sport_recognition	L04
STL-10+Image_Classification	L05	VQA_utility_affordance	L04
STVQA+image_question_answer	L09	VQARAD+question_answer	L04
textcaps+caption_generation	L10	VQAv2	L04
Total-Text+text_detection	L09	VQG+caption_generation	L02
trainSet+image_classification	L04	VQG+question_generation	L03
VIQUAE+question_answer	L04	WIKIART+art_classification	L04
VisDA-2017+image_classification	L04	wikihow_image_text_step_order	L08
VisDA-2017+object_classification_train	L05	wikihow_immediate_next_step_selection	L05
VisDA-2017+object_classification_validation	L05	wikihow_next_step	L03
visdial+answer_question_2	L01	wikihow_text_image_step_order	L08
visdial+answer_question_4	L01	Winoground+image_captioning	L05
visdial+answer_question_6	L01	WIT+detailed_description	L03
visdial+answer_question_9	L01	Yoga-82+yoga_pose_recognition	L04

APPENDIX F
COMPLETE SIGNAL-RANGE RESULTS AND DIAGNOSTIC DEFINITIONS

A. All Fifteen Selection Rules

On LLaVA-665K, each of the five signals defines a low-, middle-, or high-range rule, giving 15 rules that each select approximately 10% of the candidates. At each online selection step, the current candidates are ranked by the chosen signal as measured at that step. A contiguous block of K_t examples is selected from the bottom, center, or top of the ranking. Each rule is evaluated on the 11 metrics in Table S11.

The table contains all 165 reported scores. The names in the original score export correspond to the current notation as $r_T = d^T$, $r_V = d^V$, $g_\xi = g(\xi)$, $g_\theta = g(\theta_t)$, and $\delta_g = \delta$. For selection rule a and metric e, the rank orders scores from highest to lowest and assigns tied scores their average rank:

$$\mathrm{rank}_{a,e} = 1 + \#\{b : s_{b,e} > s_{a,e}\} + \frac{\#\{b : s_{b,e} = s_{a,e}\} - 1}{2}. \tag{S32}$$

The final column averages these ranks across the 11 metrics. The middle-range capability plot in the main paper instead ranks the five middle-range rules within each metric. The two plots therefore use different comparison sets.

Recomputation from the accompanying CSV gives mean ranks of 11.35, 5.57, and 7.08 for the low, middle, and high ranges. Middle-range selection beats low-range selection in 49 of the 55 signal–metric comparisons. The table of 15 rules also includes the six remaining comparisons and all ties. These summaries are deterministic calculations from the displayed scores.

B. Low-Range Optimization and Image Diagnostics

The optimization penalty uses midrank percentiles within each task and stage strictly below 0.1 as defined in (S17). The image analysis uses a separate diagnostic construction: for d^T and d^V, the 10th-percentile raw-signal threshold is computed among background examples grouped by scoring stage, task, and answer-length group. An image belongs to that diagnostic tail when its signal value is at or below its group's threshold. Ties at the threshold can make the observed tail rate exceed 10%.

For group g, let n_g^{bad} be the number of problematic images and b_g the fraction of background examples in the diagnostic tail. The matched background rate is

$$\bar{b} = \sum_g \frac{n_g^{\mathrm{bad}}}{\sum_u n_u^{\mathrm{bad}}} b_g. \tag{S33}$$

The thresholds and b_g values are computed separately for d^T and d^V. Among 1,056 problematic images with paired scores, their tail counts are 222 and 208, respectively, giving 21.02% and 19.70%; the matched background rate is 10.46% for each signal. The corresponding δ tail rate is 8.62%. Grouping and weighting make the background comparison match the problematic images in scoring stage, task, and answer length.

C. Exact Enrichment Denominator

For a signal range R, define $I_{i,R} = \mathbf{1}[i \text{ lies in } R]$ and selected count $O_{t,R} = \sum_{i \in S_t} I_{i,R}$. Set every candidate score to zero while retaining the stage's task quotas and clusters, and compute the resulting inclusion probabilities $\mu_{i,t}^0$. Linearity of expectation gives

$$E_{t,R}^0 = \sum_{i \in \mathcal{P}_t} I_{i,R} \mu_{i,t}^0, \qquad \mathrm{Enrich}_{t,R} = \frac{O_{t,R}}{E_{t,R}^0}, \tag{S34}$$

for $E_{t,R}^0 > 0$. A value of one matches uniform feasible selection. The range indicator and uniform baseline are evaluated on the same stage features and constraints. The plotted ranges are P80–P100 for d^T and P40–P60 for each other signal. The covariance and marginal identities in Appendix B permit this expectation to be computed from the normalizer.

Mean selected percentiles summarize the location of selected values; enrichment measures how often examples from a specified interval are selected relative to the uniform baseline. More selected examples can fall in the middle range without changing the mean. This explains why the two trajectory summaries provide complementary descriptions.

APPENDIX G
STAGE SCHEDULING AND MODALITY-GROUPING DIAGNOSTICS

The end-to-end stage-length study uses $n \in \{2, 6, 12, 18, 24\}$ at $\rho = 20\%$. It reruns online selection with the same candidate pool, initialization, total SFT updates, and evaluation protocol for each n. Stage boundaries jointly affect candidate comparisons, clustering, batch grouping, and feedback frequency. The following diagnostics from an earlier experiment examine batch grouping while holding the selected examples and model states fixed.

A. Stage Boundaries and Native Modality Grouping

The stage length n and effective global batch size B determine the full-stage selection size nB. With W devices, per-device microbatch size b, and accumulation length a, $B = baW$. The local grouping diagnostic uses one device, $b = 8$, and $a = 8$, giving $B = 64$. Thus $n = 12$ and $n = 2$ correspond to 768 and 128 examples, respectively.

In this diagnostic, the standard LLaVA modality-length sampler first separates image–text and text-only examples within the selected set. It then forms large batches, called megabatches, using dialogue lengths measured by counting words separated by whitespace. Image–text lengths are positive and text-only lengths negative. The effective device count supplied to this sampler includes gradient accumulation, giving 64 examples per megabatch in this setup. The final megabatches of the two modalities are combined into a final block ordered by the saved selection indices. Splitting 768 examples into six selections repeats this final-block operation six times. A mixed-modality update position contains both modalities across its accumulated microbatches; a mixed microbatch contains both in a single forward pass. These are distinct counts.

TABLE S11
COMPLETE REPORTED SCORES FOR THE 15 SIGNAL-RANGE RULES (APPROXIMATELY 10% ADMISSION). ALL SCORE COLUMNS ARE HIGHER-IS-BETTER; MEAN RANK IS LOWER-IS-BETTER. L-B: LLAVA-BENCH (COCO); WILD: LLAVA-BENCH-IN-THE-WILD. CSV METRIC NAMES AND PRECISION ARE PRESERVED.

Signal	Range	POPE	MME-P	MME-C	GQA	SQA	VizWiz	MMB EN	Text VQA	MM-Vet	L-B	Wild	Mean rank
δ	Low	81.57	1054.71	241.43	52.15	60.15	33.01	44.33	39.04	28.17	71.10	42.70	11.27
	Middle	84.07	1329.94	278.93	55.11	67.06	46.53	54.36	40.37	29.45	76.40	47.10	4.45
	High	83.34	1235.96	281.79	48.10	58.95	56.91	49.03	40.61	27.94	70.50	44.10	8.59
$g(\theta_t)$	Low	81.40	1088.23	286.21	48.32	65.32	41.52	23.72	32.12	31.92	69.20	39.10	10.68
	Middle	83.50	1317.15	257.14	55.61	68.57	44.72	52.38	40.43	33.62	74.50	46.90	4.91
	High	82.53	1242.67	289.24	52.67	60.15	35.34	53.69	43.47	28.21	70.50	49.30	7.18
$g(\xi)$	Low	71.53	1109.98	298.21	43.61	65.57	51.64	30.24	26.09	23.58	68.00	37.50	11.27
	Middle	82.17	1316.02	287.86	55.33	65.93	39.12	56.69	39.56	28.90	71.50	44.40	5.91
	High	81.40	1280.15	286.88	53.18	65.76	42.47	55.07	43.90	28.35	70.60	45.00	6.41
d^T	Low	81.13	1168.65	307.86	50.26	40.04	47.86	15.09	38.43	28.85	67.90	40.30	10.50
	Middle	83.20	1281.32	264.29	56.38	57.75	44.65	51.08	41.45	28.85	69.30	47.50	7.14
	High	81.57	1266.92	262.14	53.42	67.88	48.91	54.73	42.02	31.01	73.82	43.30	5.86
d^V	Low	78.60	1199.26	274.29	51.29	39.97	35.86	10.26	38.26	26.01	66.60	41.20	13.00
	Middle	83.13	1348.70	282.36	55.38	65.20	45.58	53.78	40.57	28.26	73.00	48.30	5.45
	High	82.21	1289.87	283.93	49.91	61.71	54.71	50.95	41.98	27.75	75.20	42.40	7.36

B. Fixed-Selection, Fixed-Score Replay

The replay includes all 20 stages with exactly 768 selected examples from an earlier 32-stage LLaVA-1.5-7B trajectory, totaling 15,360 sample records. The other 12 stages each contain 896 examples and do not fit the 12-position window used for this replay comparison. This trajectory was recorded before the current protocol of one PPO call per transition was adopted. The selected examples and the model scores recorded at the start of each stage stay fixed during this diagnostic.

For each selected set, $n = 12$ groups all 768 examples together; $n = 2$ randomly splits the set into six subsets of 128 and invokes the same sampler separately. The subsets are processed in the group order used in the original training run. Both settings use every example once, yielding 96 microbatches and 12 effective update positions, each corresponding to one accumulated batch. No optimizer updates occur during fixed-score replay. Grouping is repeated for 32 paired seeds: initial sampler seeds 42–73 and split seeds 1042–1073. Within each split, substage sampler seeds increment from the corresponding initial seed. These repetitions measure variation caused by random grouping.

Per-example NLL is recovered by adding the corresponding reference NLL to the stored residual; the reported maximum discrepancy from the saved scoring trace is 5.96×10^{-8}. Each example's NLL stays fixed at the value computed by the model at the start of its original stage. Replay averages losses by supervised token count within each microbatch, then gives equal weight to the eight accumulated microbatches. For Table S12, we first compute each statistic for a selected set and seed pair, then average over the 20 sets and 32 seed pairs. The standard deviation is computed over the 12 step losses.

TABLE S12
FIXED-SCORE REPLAY WITH THE SAME EXAMPLES AND 12 EFFECTIVE UPDATE POSITIONS PER WINDOW.

Statistic	$n = 12$	$n = 2$
Mixed-modality positions per 12	1.000	6.000
Mixed microbatches per 96	0.800	4.900
Minimum step loss per window	0.459683	0.698419
Within-window loss std. dev.	0.154145	0.119582
Mean step loss	0.948467	0.885424
Positions with loss ≤ 0.4 per window	0.131250	0.000000

Smaller stages spread low-loss examples across more positions. Under the replay averaging scheme, this raises the minimum step loss while lowering the mean. Both changes come from regrouping the same fixed scores. Figure S1 shows all positions for sampler seed 42 and split seed 1042; Fig. S2 shows the final five eligible selections in historical order. Each 12-position block uses the scores recorded at the start of its original stage. The plot places these blocks in sequence to show the fixed-score replay.

C. Grouping and Effective Gradient Weights

Let $\ell_i(\theta)$ be example i's mean supervised-token NLL, ν_i its supervised token count, $\mathcal{U}_m$ microbatch m, and $T_m = \sum_{i \in \mathcal{U}_m} \nu_i$. Native SFT averages supervised token losses within each microbatch, then weights a accumulated microbatches equally. For one device,

$$L_{\text{step}}(\theta) = \frac{1}{a} \sum_{m=1}^{a} \frac{\sum_{i \in \mathcal{U}_m} \nu_i \ell_i(\theta)}{T_m}. \tag{S35}$$

TABLE S13
LOSSES FROM FORWARD PASSES USING THE STANDARD IMPLEMENTATION ON THE SAME 16 EXAMPLES. THE MEAN WEIGHTS THE TWO MICROBATCHES EQUALLY.

Grouping	Batch 1	Batch 2	Mean
Separate	0.334435	1.224146	0.779290
Mixed	0.619250	0.488524	0.553887

Ignoring numerical error and assuming that examples are processed independently, the gradient at fixed model parameters is

$$\nabla_\theta L_{\text{step}} = \sum_{m=1}^{a} \sum_{i \in \mathcal{U}_m} \underbrace{\frac{\nu_i}{a T_m}}_{w_i} \nabla_\theta \ell_i. \tag{S36}$$

Changing microbatch membership changes T_m, and therefore the sample coefficients w_i. If the sample gradients do not all lie along the same line, changing these weights can change the direction of the total gradient. If $T_{m,\text{text}}$ counts text-only supervised tokens in microbatch m, the effective text weight is

$$W_{\text{text}} = \sum_{i \in \text{text}} w_i = \frac{1}{a} \sum_{m=1}^{a} \frac{T_{m,\text{text}}}{T_m}. \tag{S37}$$

For sampler/split seeds 42/1042, the mean over all 240 positions is 6.1811% for $n = 12$ and 7.0315% for $n = 2$. The numbers of examples and tokens are identical, but normalizing within each microbatch changes the weight of text-only examples in the objective. This sum describes their weight in the loss. Their effect on the gradient also depends on the directions and magnitudes of individual gradients.

D. Forward-Pass and Gradient Measurements

At a fixed LLaVA-1.5-7B reference checkpoint, the local test takes the first eight text-only and eight image–text examples in the saved order of historical selection 28. Separate grouping places the eight examples of each modality into their own microbatch. Mixed grouping forms two microbatches, each with four text-only and four image–text examples. Both settings use the same 16 examples and parameters, accumulate two microbatches, and perform no optimizer step.

Supervised token counts for each example are unchanged. The maximum per-example NLL change is 0.002757; losses from the standard implementation agree with independently recomputed token means to within 2×10^{-6}. The separate microbatches contain 6,404 and 1,932 supervised tokens; mixed microbatches contain 3,335 and 5,001. Under (S36), regrouping multiplies the per-token coefficients of text-only examples by 1.9202 or 1.2805, and those of image–text examples by 0.5793 or 0.3863.

We run backward passes on these four microbatches to compute gradients for the language model's LoRA parameters. This diagnostic checkpoint has rank 128 and 319,815,680 trainable parameters. The base model, vision tower, and projector are frozen; dropout is disabled, and both groupings use the same gradient-checkpointing setting. This fixed-checkpoint diagnostic is separate from the main training runs. For separate and mixed gradients $\mathbf{g}_{\text{sep}}$ and $\mathbf{g}_{\text{mix}}$, the relative difference is $\|\mathbf{g}_{\text{mix}} - \mathbf{g}_{\text{sep}}\|_2 / \|\mathbf{g}_{\text{sep}}\|_2$.

TABLE S14
LOCAL GRADIENT COMPARISON USING EQUAL MICROBATCH WEIGHTS OR A GLOBAL TOKEN MEAN.

Aggregation	Cosine	Relative diff.
Equal microbatch weights	0.881261	52.5921%
Global supervised-token mean	0.999933	1.1638%

Under native aggregation, the gradient norms are 0.341052 and 0.379230 for separate and mixed grouping. The control reweights the same four microbatch gradients by their supervised-token counts. Both groupings then compute the global token mean over all 16 examples. The resulting gradients are nearly aligned, supporting the interpretation that normalization denominators explain most of the local difference.

These diagnostics show how stage boundaries change grouping and, in turn, the weights used in optimization. Fixed-score replay holds the selected samples and model scores constant, and the gradient test measures the local effect at a fixed parameter state. The end-to-end stage-length study evaluates the combined effect of candidate comparisons, clustering, batching, and feedback frequency.

APPENDIX H
EVALUATION DEFINITIONS AND AGGREGATE SENSITIVITY

A. Included Metrics and Normalization

For benchmark set $\mathcal{E}$, method score $a_{m,e}$, and full-data score $a_{\text{full},e}$, relative performance and minimum retention are

$$\begin{aligned} \text{Rel.}(m) &= \frac{100\%}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \frac{a_{m,e}}{a_{\text{full},e}}, \\ \text{MinRet}(m) &= 100\% \min_{e \in \mathcal{E}} \frac{a_{m,e}}{a_{\text{full},e}}. \end{aligned} \tag{S38}$$

Every included metric has equal weight after division by its own full-data score. Each dataset–model setting uses its own reference. Image-replacement experiments retain the clean full-data denominator at each corruption rate.

MME-P and MME-C denote perception and cognition totals. The 13B table uses the combined MME score as one metric. LLaVA-W and LLaVA Wild are names for the same benchmark in the respective tables. The signal-range study in Table S11 has its own 11-metric set, including LLaVA-Bench (COCO). Its mean ranks use the metric set displayed in that table.

For the main comparison tables, bold and underlining identify the best and second-best distinct values among methods at 20% selection, including ties and the transfer variant. Full-data scores supply the reference denominator. Rank counts cover benchmark columns and exclude the aggregate Rel. column.

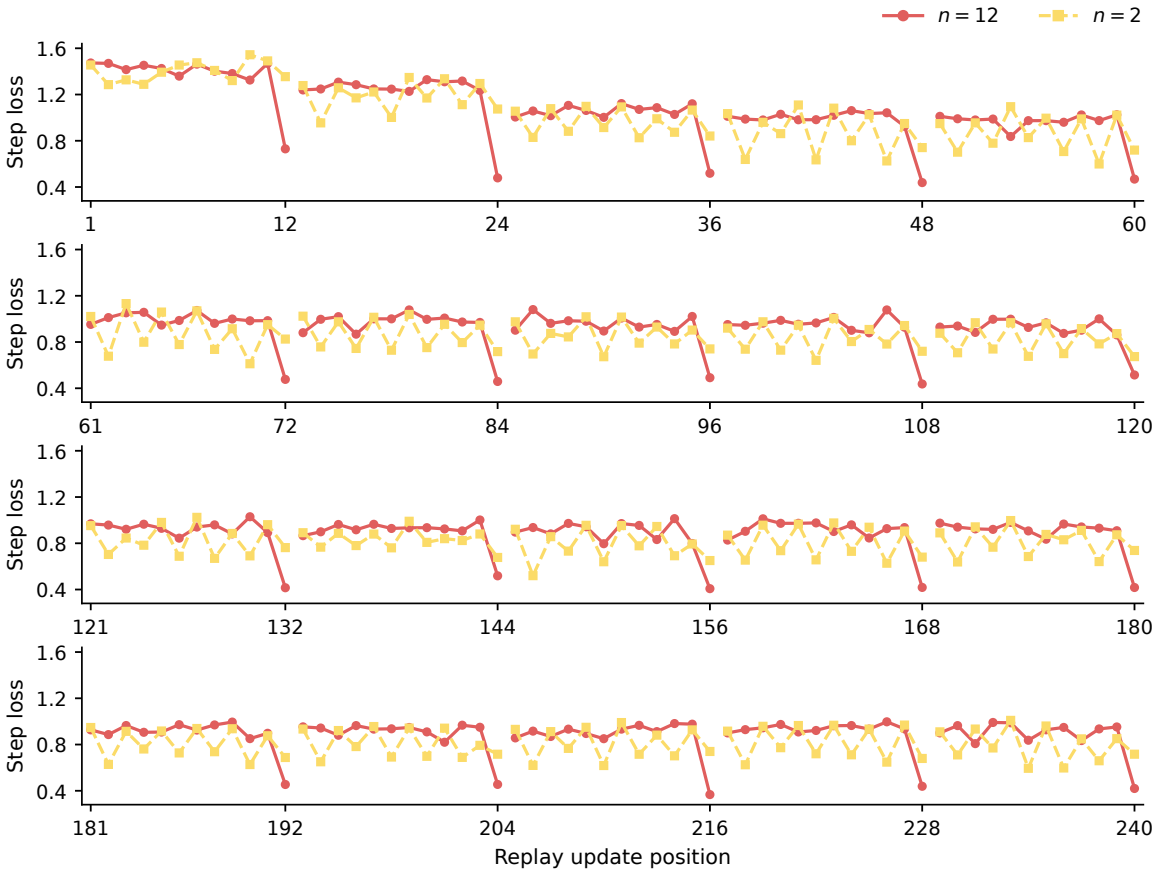


Fig. S1. All 240 fixed-score replay positions from 20 eligible historical selections, with sampler seed 42 and split seed 1042. Each row shows 60 positions on the same vertical scale. Each 12-position block uses one selected set and the model scores from the start of that stage; curves are disconnected between blocks. Summary statistics over 32 seed pairs appear in Table S12.

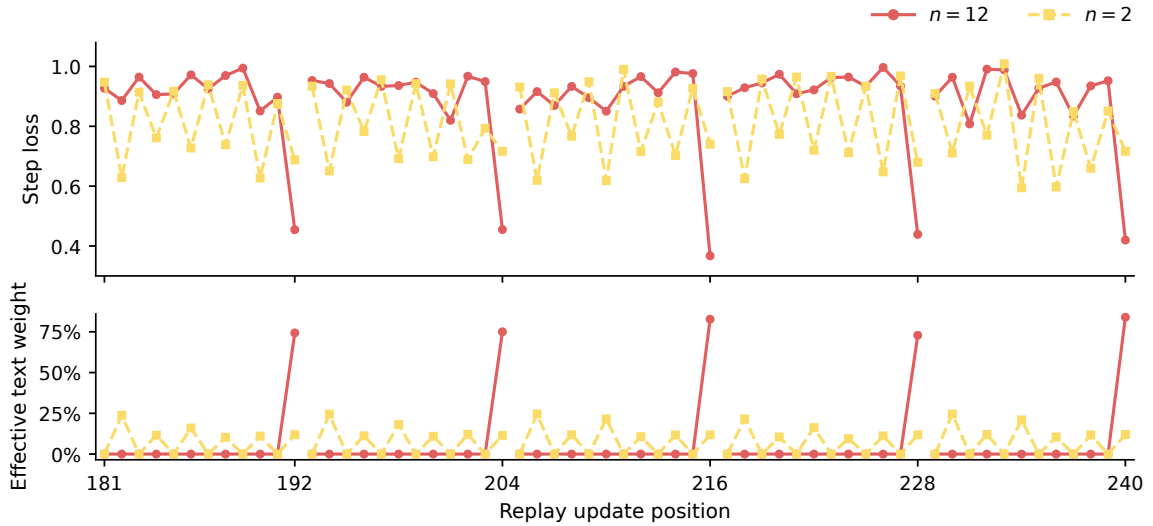


Fig. S2. Final 60 positions of the same replay as Fig. S1. Top: fixed-score step loss. Bottom: effective text weight from (S37). The five windows are the final eligible selections in historical order.

TABLE S15
METRICS INCLUDED IN THE THREE MAIN RESULT TABLES.

Setting	Count	Included metrics
LLaVA-665K, 7B	12	POPE, MME-P, MME-C, SQA-I, GQA, VizWiz, VQAv2, MMB-en, MMB-cn, TextVQA, MM-Vet, LLaVA Wild
Vision-Flan, 7B	11	VQAv2, GQA, LLaVA Wild, VizWiz, SQA-I, TextVQA, MME-P, MMB-en, MMB-cn, POPE, MM-Vet
LLaVA-665K, 13B	10	VQAv2, GQA, VizWiz, SQA-I, TextVQA, POPE, MME, MMB-en, MMB-cn, LLaVA-W

TABLE S16
COMPLETE METRIC-OMISSION ANALYSIS. EACH PAIR OF COLUMNS GIVES AN OMITTED METRIC AND THE RESULTING ORIENT-MINUS-COIDO GAP IN PERCENTAGE POINTS.

LLaVA-665K, 7B		Vision-Flan, 7B		LLaVA-665K, 13B	
Omitted metric	Gap	Omitted metric	Gap	Omitted metric	Gap
POPE	1.135	VQAv2	1.159	VQAv2	1.147
MME-P	1.054	GQA	0.856	GQA	1.290
MME-C	1.258	LLaVA Wild	0.844	VizWiz	0.412
SQA-I	1.280	VizWiz	0.962	SQA-I	1.358
GQA	1.317	SQA-I	1.326	TextVQA	1.165
VizWiz	0.686	TextVQA	1.024	POPE	1.151
VQAv2	1.233	MME-P	1.157	MME	1.262
MMB-en	1.242	MMB-en	1.431	MMB-en	1.008
MMB-cn	1.311	MMB-cn	1.046	MMB-cn	0.968
TextVQA	1.219	POPE	1.167	LLaVA-W	1.058
MM-Vet	0.644	MM-Vet	0.349	–	–
LLaVA Wild	0.681	–	–	–	–

TABLE S17
AGGREGATE GAP AND RANGE OF GAPS AFTER OMITTING ONE METRIC. ALL VALUES ARE PERCENTAGE POINTS.

Setting	All metrics	Min.	Max.
LLaVA-665K, 7B	1.088	0.644	1.317
Vision-Flan, 7B	1.029	0.349	1.431
LLaVA-665K, 13B	1.082	0.412	1.358

B. Removing Each Metric in Turn

Let $q = |\mathcal{E}|$ and let e be the omitted metric. Holding every full-data denominator fixed, recompute

$$\mathrm{Rel.}_{-e}(m) = \frac{100\%}{q-1} \sum_{u \in \mathcal{E}\setminus\{e\}} \frac{a_{m,u}}{a_{\mathrm{full},u}}. \tag{S39}$$

The paired difference is $\Delta_{-e} = \mathrm{Rel.}_{-e}(\textsc{Orient}) - \mathrm{Rel.}_{-e}(\textsc{CoIDO})$, in percentage points. Tables S17 and S16 use the rounded per-metric scores displayed in the current manuscript. The accompanying CSV preserves those inputs, and the analysis script recomputes all 33 omissions.

All 33 gaps are positive. The smallest gaps arise after omitting MM-Vet in the two 7B settings and VizWiz in the 13B setting. Rounded to two decimals, they are 0.64, 0.35, and 0.41 percentage points. This analysis shows how much each benchmark affects the aggregate comparison.

APPENDIX I
COMPANION FILES AND EXPERIMENT RECORDS

A. Repository and Companion Contents

The code repository is github.com/Meltcjm/ORIENT. This companion includes its standalone LaTeX sources, two historical replay figures, the complete signal-range CSV, the exported main-comparison scores used to analyze metric omissions, and the 191-task mapping used in the transfer experiment (Appendix E-B). Its scripts rebuild the signal-range and task-mapping tables, recompute all metric omissions, and check the inference identities by enumerating all feasible subsets in small test cases. The mapping JSON and CSV include examples that explain the mapping decisions; a separate adapter validates the task files and binds source identities to checkpoint coordinates.

The grouping and gradient measurements in Appendix G follow the historical experimental record supplied with the replay figures. Their sample counts, seed ranges, checkpoint scope, and aggregation formulas are stated in that section. The SFT reference script and the global batch-size adjustment for two devices are provided with the companion configurations.

B. Implementation Conventions

The implementation snapshot inspected for the numerical discussion is identified by the abbreviated commit ID `5f2cbecbc6d8`; its full identifier and module paths are recorded in the companion README. In this snapshot, the Critic's early history uses the available preceding rewards, up to eight. Before any completed stage, its preceding-reward, mean, and standard-deviation entries are zero. The standard

deviation divides by the number of available rewards, following the population formula. Ties between quota remainders are broken using a task order rotated by the stage index. Ties between nearest neighbors are broken using the supplied candidate order. These details specify this code snapshot's deterministic conventions.

C. Execution Records

Each stage record lists the stage index, candidate count, selected count and indices, and cumulative optimizer step. It also includes the feedback losses before and after SFT, per-task feedback losses, and reward. The policy update is attached to the transition that produced that reward. The final summary records completed stages, selected count, optimizer updates, final feedback loss, and reward sum.

To reproduce an experiment, these records need to be accompanied by the candidate/feedback split, source annotations, model and tokenizer versions, trainable-parameter settings, evaluation command, and hardware timing. Transfer also needs the mapping from native tasks to source identities. Image-replacement runs require a record of the swapped image pairs and an identifier for the score cache computed from the corrupted inputs. Together these identifiers connect the method specification to each reported result.

References

[1] Z. Xu, C. Feng, R. Shao, T. Ashby, Y. Shen, D. Jin, Y. Cheng, Q. Wang, and L. Huang, "Vision-Flan: Scaling human-labeled tasks in visual instruction tuning," in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 15 271–15 342.

[2] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 26 296–26 306.

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